Additional Exercise 7 Practice Questions

Bisecting K-means and DBSCAN

1 Problem 1: Bisecting K-means practice

Execute bisecting K-means with $K = 3$ leaf clusters using Euclidean distance on the following data. In each ordinary K-means split, use $k = 2$.

ID	P1	P2	P3	P4	P5	P6
x	1.0	1.0	2.0	6.0	7.0	9.0
y	1.0	2.0	1.0	6.0	6.0	1.0

Use this deterministic setup:

- First bisection initial centroids: $P1 = (1, 1)$ and $P6 = (9, 1)$.
- For the second bisection, use initial centroids $P4 = (6, 6)$ and $P6 = (9, 1)$.

For every iteration, report:

1. the assignment of each point;
2. the updated centroid positions;

2 Problem 2: DBSCAN practice

Cluster the points below using DBSCAN with:

$$\varepsilon = 1.0, \qquad minPts = 3.$$

Point	x	y
A	1.0	1.0
B	1.4	1.0
C	1.0	1.4
D	2.2	1.1
E	5.0	5.0
F	5.8	5.1

Tasks:

1. Determine the ε-neighborhood of every point.
2. Label every point as core, border, or noise.
3. Give the resulting DBSCAN clusters.
4. Quick extension: what would change if $minPts = 2$ while ε stays 1.0?

3 Solution 1: Bisecting K-means practice

3.1 First bisection, iteration 1

Initial centroids:

$$\mu_A = P1 = (1, 1), \qquad \mu_B = P6 = (9, 1).$$

Point	d^2 to $(1, 1)$	d^2 to $(9, 1)$	Assignment
P1	0.00	64.00	C_A
P2	1.00	65.00	C_A
P3	1.00	49.00	C_A
P4	50.00	34.00	C_B
P5	61.00	29.00	C_B
P6	64.00	0.00	C_B

Assignments:

$$C_A = \{P1, P2, P3\}, \qquad C_B = \{P4, P5, P6\}.$$

Updated centroid for C_A:

$$\mu_A = \left(\frac{1+1+2}{3}, \frac{1+2+1}{3}\right) = \left(\frac{4}{3}, \frac{4}{3}\right) = (1.3333, 1.3333).$$

Updated centroid for C_B:

$$\mu_B = \left(\frac{6+7+9}{3}, \frac{6+6+1}{3}\right) = \left(\frac{22}{3}, \frac{13}{3}\right) = (7.3333, 4.3333).$$

SSE:

$$SSE(C_A) = 1.3333,$$
$$SSE(C_B) = 21.3333,$$
$$SSE_{total} = 22.6666.$$

3.2 First bisection, iteration 2

Current centroids:

$$\mu_A = (1.3333, 1.3333), \qquad \mu_B = (7.3333, 4.3333).$$

Point	d^2 to $(1.3333, 1.3333)$	d^2 to $(7.3333, 4.3333)$	Assignment
P1	0.2222	51.2222	C_A
P2	0.5556	45.5556	C_A
P3	0.5556	39.5556	C_A
P4	43.5556	4.5556	C_B
P5	53.8889	2.8889	C_B
P6	58.8889	13.8889	C_B

Assignments do not change, so the first bisection has converged.
After the first bisection:

$$C_A = \{P1, P2, P3\}, \qquad SSE = 1.3333,$$
$$C_B = \{P4, P5, P6\}, \qquad SSE = 21.3333.$$

The larger SSE cluster is C_B, so split $\{P4, P5, P6\}$ next.

3.3 Second bisection, iteration 1

Initial centroids for the second bisection:

$$\mu_U = P4 = (6,6), \qquad \mu_V = P6 = (9,1).$$

Point	d^2 to $(6,6)$	d^2 to $(9,1)$	Assignment
P4	0.00	34.00	C_U
P5	1.00	29.00	C_U
P6	34.00	0.00	C_V

Assignments:

$$C_U = \{P4, P5\}, \qquad C_V = \{P6\}.$$

Updated centroids:

$$\mu_U = \left(\frac{6+7}{2}, \frac{6+6}{2}\right) = (6.5, 6),$$
$$\mu_V = (9,1).$$

SSE:

$$SSE(C_U) = 0.5,$$
$$SSE(C_V) = 0.$$
$$SSE_{split} = 0.5.$$

3.4 Second bisection, iteration 2

Current centroids:

$$\mu_U = (6.5, 6), \qquad \mu_V = (9,1).$$

Point	d^2 to $(6.5,6)$	d^2 to $(9,1)$	Assignment
P4	0.25	34.00	C_U
P5	0.25	29.00	C_U
P6	31.25	0.00	C_V

Assignments do not change, so the second bisection has converged.

3.5 Final answer for Problem 1

Final cluster	Points	Centroid	SSE
1	$\{P1, P2, P3\}$	$(1.3333, 1.3333)$	1.3333
2	$\{P4, P5\}$	$(6.5, 6)$	0.5000
3	$\{P6\}$	$(9,1)$	0.0000

$$SSE_{final} = 1.3333 + 0.5000 + 0.0000 = 1.8333.$$

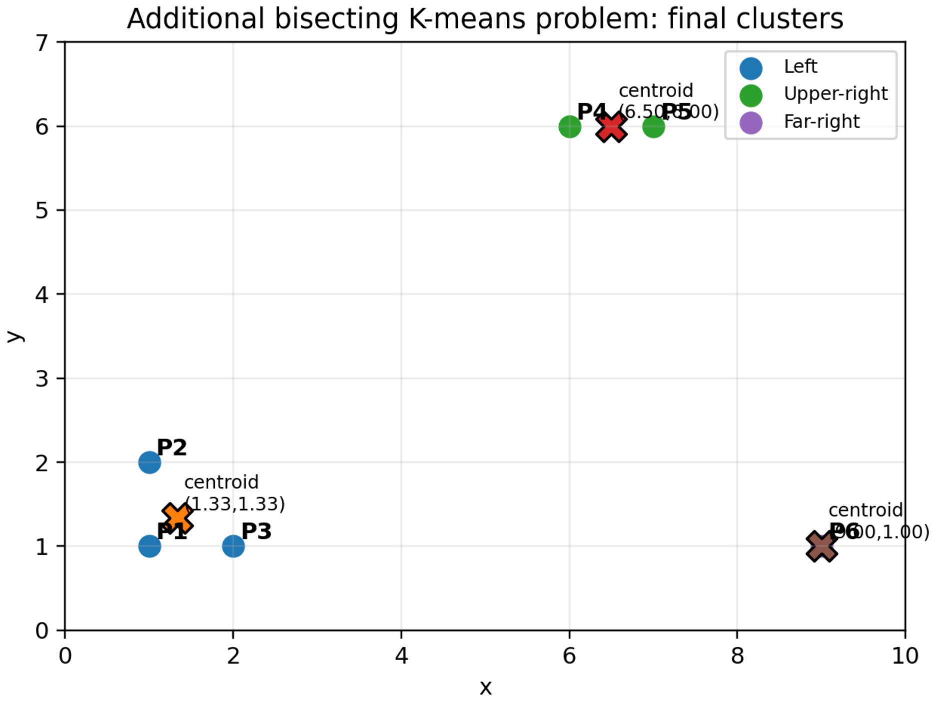

Figure 1: Final clusters for the additional bisecting K-means problem.

4 Solution 2: DBSCAN practice

4.1 Distances that matter

Only distances less than or equal to $\varepsilon = 1.0$ matter.

$$d(A, B) = \sqrt{(1 - 1.4)^2 + (1 - 1)^2} = 0.4.$$

$$d(A, C) = \sqrt{(1 - 1)^2 + (1 - 1.4)^2} = 0.4.$$

$$d(B, C) = \sqrt{(1.4 - 1)^2 + (1 - 1.4)^2} = \sqrt{0.32} = 0.5657.$$

$$d(B, D) = \sqrt{(1.4 - 2.2)^2 + (1 - 1.1)^2} = \sqrt{0.65} = 0.8062.$$

$$d(E, F) = \sqrt{(5 - 5.8)^2 + (5 - 5.1)^2} = \sqrt{0.65} = 0.8062.$$

The other pairwise distances are greater than 1.0.

4.2 Neighborhoods and point types

Point	$N_\varepsilon(p)$	Count	Type
A	$\{A, B, C\}$	3	core
B	$\{A, B, C, D\}$	4	core
C	$\{A, B, C\}$	3	core
D	$\{B, D\}$	2	border
E	$\{E, F\}$	2	noise
F	$\{E, F\}$	2	noise

Explanation:

- A, B, and C are core because each has at least 3 points in its ε-neighborhood.
- D is not core because it has only 2 points in its neighborhood, but it is within ε of B, which is core. Therefore D is border.
- E and F are close to each other, but neither is core and neither is close to a core point. Therefore both are noise.

4.3 Final DBSCAN clusters

Core component:

$$\{A, B, C\}.$$

Border point D attaches through core point B. Therefore:

$$\boxed{Cluster\ 1 = \{A, B, C, D\}}.$$

Noise:

$$\boxed{Noise = \{E, F\}}.$$

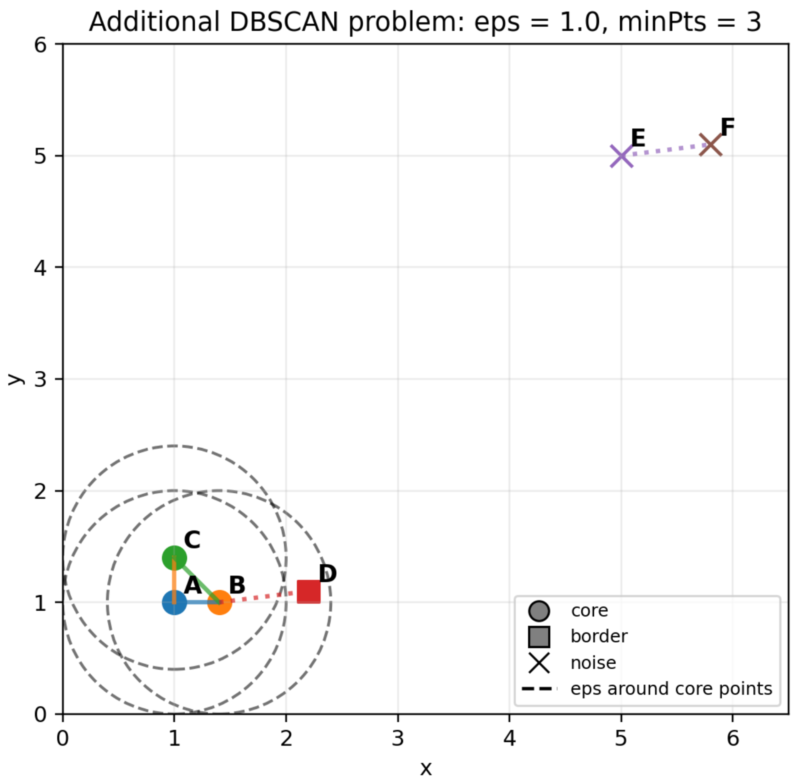

Figure 2: Solution visualization. A, B, and C are core. D is border. E and F are noise under $minPts = 3$.

4.4 Quick extension answer

If $minPts = 2$ while $\varepsilon = 1.0$ stays the same, then E and F would each have neighborhood size 2:

$$N_\varepsilon(E) = \{E, F\}, \qquad N_\varepsilon(F) = \{E, F\}.$$

With $minPts = 2$, both E and F become core. Then $\{E, F\}$ becomes a second DBSCAN cluster.

So the result would change to:

$$Cluster\ 1 = \{A, B, C, D\}, \qquad Cluster\ 2 = \{E, F\}.$$

This is useful for teaching because it shows that two close points are not automatically a cluster. Whether they become a cluster depends on the density threshold.